

Abstract No.

CIITA-mediated MHC-II expression in glioblastoma unveil MHC-II-bound tumor-associated peptides for protective anti-tumor vaccination

Mariam Shallak¹, Mattia Volpi^{1,*}, Sofia Forner¹, Michal Bassani-Sternberg², Roberto S. Accolla¹, and Greta Forlani¹

¹Laboratories of General Pathology and Immunology "Giovanna Tosi"

²University of Laussane, Ludwig Cancer Research Center, Laussane, Switzerland

*Presenting author

Glioblastoma multiforme (GBM) is the most aggressive primary brain tumor in adults and remains largely incurable. Peptide vaccination is a promising immunotherapeutic strategy, but most clinical approaches target MHC-I-restricted antigens to stimulate CD8⁺ T cells, with limited benefit. We previously showed that CIITA-mediated MHC-II expression in GL261 glioblastoma cells induces tumor rejection or growth retardation and durable CD4⁺ T-cell-mediated protection.

Here, we used high-throughput immunopeptidomics to characterize MHC-II ligands from murine glioma cell lines (GL261, CT2A, SB28) and their CIITA-transfected counterparts. CIITA robustly induced MHC-II surface expression, enabling identification of over 8,000 unique MHC-II-bound peptides per CIITA-expressing line. Comparative analysis identified a shared core of 5,559 peptides from 1,109 source proteins across all models. Cross-species comparison of GL261-CIITA with CIITA-expressing human GBM cell lines identified 225 conserved source proteins. Five glioblastoma-associated proteins were prioritized, yielding seven candidate MHC-II peptides for immunogenicity testing. C57BL/6 mice immunized with the peptide pool plus CpG adjuvant developed peptide-specific *ex vivo* IFN- γ responses, with peptides 1, 2, and 4 showing strongest immunogenicity. In an orthotopic GL261 model, these peptides induced antigen-specific antitumor immunity. Complete protection was observed in 30%, 40%, and 20% of mice vaccinated with peptides 1, 2, and 4, respectively. At six months after tumor implantation, survival was 50% (5/10), 30% (3/10), and 20% (2/10), respectively; all CpG-only controls died by day 23. In tumor-bearing mice, peptides 2 and 4 significantly reduced tumor size versus CpG alone, a nonimmunogenic control peptide, or an unrelated immunogenic peptide. All immunogenic peptides increased intratumoral CD3⁺ T-cell infiltration; peptide 2 increased CD4⁺ infiltration, while peptides 2 and 4 promoted CD8⁺ recruitment. Peptide 2 protection required both T-cell subsets, as *in vivo* depletion of either CD4⁺ or CD8⁺ T cells abolished efficacy. These findings identify conserved glioma-associated MHC-II peptides and support CD4⁺ T-cell-targeted peptide vaccination for GBM.



Supported by SciSerTec

Web: <http://www.scisertec.de>